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Stroke

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ORIGINAL RESEARCH ARTICLE

Change in Thigh Muscle Cross-Sectional Area Through Administration of an Anabolic Steroid During Routine Stroke Rehabilitation in Hemiplegic Patients

ABSTRACT

Okamoto S, Sonoda S, Tanino G, Tomida K, Okazaki H, Kondo I: Change in thigh muscle cross-sectional area through administration of an anabolic steroid during routine stroke rehabilitation in hemiplegic patients. *Am J Phys Med Rehabil* 2011;90:106–111.

Objective: The aim of this study was to clarify the effect of administration of an anabolic steroid (AS) without the addition of specific training in stroke patients by measuring the cross-sectional area (CSA) of the thigh.

Design: Twenty-six hemiplegic stroke patients during subacute rehabilitation were randomly assigned to a metenolone enanthate (ME) administration group or a control group (CT group). In the ME group, ME (100 mg) was injected intramuscularly weekly for 6 wks in the ME group. The CSA of the bilateral thigh muscles was measured using computed tomography. Motor subscore of the Functional Independence Measure (FIM-M) was assessed before the experimental period.

Results: At the end of 6 wks, the CSA increase in the ME group (13.4%, affected side; 14.5%, unaffected side) was significantly larger than that in the CT group (3.3%, affected side; 5.2%, unaffected side). Correlation coefficients between the initial FIM-M score and the CSA increase at 6 wks were -0.754 for the affected side and -0.567 for the unaffected side in the ME group and 0.199 for the affected side and 0.431 for the unaffected side in the CT group.

Conclusions: ME administration is effective for improving muscle CSA and, thus, muscle strengthening in stroke rehabilitation. The CSA increase in the ME group was most prominent in patients with a low initial FIM-M score.

Key Words: Anabolic Steroid, Cerebrovascular Disorders, Muscles, Activities of Daily Living

Improvement in activities of daily living (ADL) via strengthening of the muscles of the limbs in immobilized patients is a critical issue in rehabilitation. However, a severe degree of paralysis, decreased initiative, decreased cardiovascular fitness, or the presence of comorbidities that constrain patients to be at rest often prevent muscle-strengthening exercise in such patients.

Metenolone enanthate (ME) is one of the anabolic steroids (ASs), a class of medications that has a muscle-strengthening effect exceeding that of testosterone and is used to treat conditions such as osteoporosis and anemias resulting from deficient red cell production. ASs have been used in competitive sports as doping agents that assist muscle hypertrophy and increase muscle strength. The mechanism of the effect of AS consists of activation of satellite cells followed by an increase in the number of muscle fibers^{1,2} and stimulation of the growth hormone insulin-like growth factor-1 axis.³ However, many unsolved issues surround the mechanism of the effect of AS, such as the mechanism that triggers their effect.^{1,2}

In a clinical application of testosterone, increases in grip strength and hamstring muscle strength were reported,⁴ although failure to increase muscle strength was also reported.^{5,6} Therefore, the effectiveness of testosterone for muscle strengthening is controversial. An increase in muscle strength through AS administration in combination with muscle-strengthening training in patients with stroke or in a deconditioned state was reported.^{7,8} We aimed to clarify the effect of ME during the subacute rehabilitation phase in hemiplegic stroke patients.

In clinical situations, measurements of muscle torque might be inaccurate in patients with disturbance of consciousness or dementia. Decreased initiative or an uncooperative attitude might result in a mistakenly low value for muscle strength. Therefore, we used not muscle torque but measurement of the muscle cross-sectional area (CSA). Generally speaking, the CSA correlates with muscle torque,^{9,10} and we confirmed that this relationship existed in stroke patients.¹¹

SUBJECTS AND METHODS

Subjects

Subjects were 26 hemiplegic stroke patients (18 men and 8 women; mean $\pm$ SD age, 65.3 $\pm$ 13.3 yrs) who were admitted to the rehabilitation wards of our hospital from October 2004 to November 2009. These were classified as subacute

rehabilitation wards, which are designated as “kaifukuki rehabilitation wards” in Japan. Patients who had severe sensory disturbance, aphasia, cognitive dysfunction, diabetes mellitus, or liver dysfunction were excluded as subjects. All patients who met the criteria and agreed to join this trial were randomly assigned to an ME administration group (ME group) or a control group (CT group). Before assignment to either the ME or the CT group, all study participants were given information about the aim of this study and on the possible effects of ME and signed an informed consent agreement. The ethics committee at Fujita Health University Nanakuri Sanatorium reviewed and approved this research protocol (no. 3).

Methods

The experimental period was set as 6 wks. ME (100 mg) was intramuscularly injected once a week for 6 wks in the ME group after confirming that the prostate-specific antigen level was within the reference range in male patients. Every 2 wks, blood samples were obtained from patients in the ME group to examine liver function and blood sugar levels.

Rehabilitation as Background Activity for This Research:

In general, in Japan, after intensive treatment in an acute care hospital, disabled patients with stroke, traumatic brain injury, spinal cord injury, and other disabling conditions are transferred to a kaifukuki rehabilitation ward. The medical insurance system specifies that admission to a kaifukuki rehabilitation ward must take place within 2 mos of stroke onset; those who remain in an acute care facility longer are then transferred to geriatric hospitals or extended care facilities.

As is routine in kaifukuki rehabilitation wards, both the ME and CT groups participated in the full-time integrated treatment program¹² every day during their admission. The full-time integrated treatment program consists of daily rehabilitation, active life in the ward, and close communication of patient information among staff. Patients are encouraged to be active during the daytime using the 6-m-wide corridor between patient rooms and a gym. Regular rehabilitation treatment is composed of 40–60 mins of physical therapy and 40–60 mins of occupational therapy per day.

Measurements of CSA:

The CSA of the bilateral thigh muscles in both groups was measured using computed tomography (Light Speed4; GE Yokogawa Medical, Inc, Tokyo, Japan) before the experimental period and at the

second, fourth, and sixth weeks of the study period. With the patient in the supine position, transverse scans were performed by helical computed tomography at the midpoint between the great trochanter and the upper edge of the patella. To identify the muscle area, the area encompassing knee flexors, knee extensors, and hip adductors, including the femoral bone, was drawn on the computer screen using a mouse; thereafter, the computer calculated that area. A closed curved line was drawn by hand on the computer screen that displayed the cross-section of the thigh. We intended to include thigh muscles, and inevitably, the femoral bone and subcutaneous tissues were also included in this region. A histogram was drawn based on the computed tomographic value for the region under study. Then subcutaneous tissues and bone were excluded according to the difference in computed tomographic values obtained by computer calculations. Serial changes in the CSA were expressed in percentages. The researcher who measured the CSA was blinded to whether the patient was in the ME or the CT group.

Measurements of Daily Activity:

The ADL score was used as an index of daily activity. The motor subscore of the Functional Independence Measure (FIM-M)¹³ was assessed at the beginning of the study period in both groups, as was the ADL score. We felt that it was unnecessary to assess the cognitive subscore of the FIM because patients with poor cognition who could not obey instructions were excluded from the study.

Statistical Analysis:

Values for the CSA were compared between the ME group and the CT group. Statistical assessments were performed by the Wilcoxon's signed-rank test and Mann-Whitney *U* test using a statistical software (JMP8; SAS Institute Inc, Cary, NC). The level of significance was set as 0.05. The correlation

between the initial FIM-M score and the rate of increase in the CSA was calculated. Multiple regression analysis was performed for each group in which the dependent variable was the CSA increase and independent variables were FIM-M score and age.

RESULTS

Table 1 shows the clinical and demographic data of the study subjects. Ten subjects had a cerebral infarction and 16 subjects had a cerebral hemorrhage. There were 15 patients in the ME group and 11 patients in the CT group. Age and the initial FIM-M score did not differ between the two groups. Mean $\pm$ SD days from onset to admission to the rehabilitation ward was 37.6 ± 15.2 . There was a significant difference in the days from onset to admission ($P = 0.048$), but there was no significant difference in the start of experimental period from stroke onset. No patient in the ME group dropped out during the series of six injections. Adverse responses in the ME group were an increased fasting blood sugar level (two cases), hyperproteinemia (five cases), and increased cholinesterase level (three cases). Mild liver dysfunction was observed in three cases. None of these events was sufficiently serious to stop ME administration. These conditions all resolved after termination of ME administration.

Mean $\pm$ SD initial CSA was $86.8 \pm 31.3 \text{ cm}^2$ and $105.3 \pm 34.7 \text{ cm}^2$ for the affected side and unaffected side, respectively, in the ME group and $81.3 \pm 20.4 \text{ cm}^2$ and $89.6 \pm 25.0 \text{ cm}^2$, respectively, in the CT group, with no significant difference shown by the Wilcoxon's signed-rank test between the two groups. Figure 1 shows the serial changes in the CSA, with significant differences ($P < 0.05$) indicated between various time points as shown in Table 2. The bilateral CSA increased in both groups. The difference in the CSA between the two groups, determined by subtracting the average CSA in the

TABLE 1 Data on enrolled patients

	Metenolone Enanthate Group (n = 15)	Control Group (n = 11)
Age, yrs	62.3 $\pm$ 14.3	69.4 $\pm$ 11.1
Sex (male/female)	12/3	6/5
Type of stroke (infarct/hemorrhage)	7/8	3/8
Hemiplegic side (right/left)	8/7	3/8
Days from onset to admission	41.1 $\pm$ 11.1	31.4 $\pm$ 11.2
Start of experimental period from stroke onset, days	45.9 $\pm$ 16.6	42.3 $\pm$ 14.1
Initial motor subscore of the Functional Independence Measure	42.1 $\pm$ 15.1	43.4 $\pm$ 16.6

Values are presented as mean $\pm$ SD or number of patients.

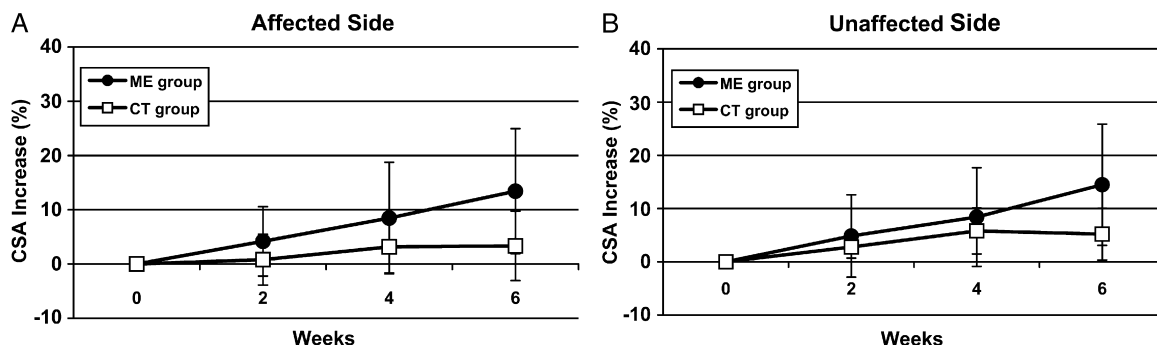


FIGURE 1 A–B, Serial changes in the cross-sectional area (CSA) of the thigh muscle. Serial change in the CSA of the thigh muscle is expressed by the percentage increase in both the (A) affected and (B) unaffected sides in the metenolone enanthate (ME) group and the control (CT) group. Significant differences ($P < 0.05$) between various time points or between the ME and CT groups are shown in Table 2. The difference in CSA increase became significant between the ME and CT groups at 6 wks.

CT group from that in the ME group, tended to become larger over time. At the sixth week of the study period, the increase in the CSA in the ME group (mean $\pm$ SD, 13.4% $\pm$ 11.5%, affected side; 14.5% $\pm$ 11.4%, unaffected side) was significantly larger than that in the CT group (3.3% $\pm$ 6.4%, affected side; 5.2% $\pm$ 4.9%, unaffected side).

Figure 2 shows the relationship between the initial FIM-M score and the CSA increase at 6 wks. The lower the initial FIM-M was, the greater the CSA increase. The correlation coefficient was -0.754 and -0.567 in the ME group and 0.199 and 0.431 in the CT group for the affected side and unaffected side, respectively.

The multiple regression equation for calculating the CSA increase at 6 wks derived from data on the ME group was as follows:

$$\text{CSA increase} = 30.9 - 0.423 \times \text{FIM-M} + 0.022 \times \text{age}$$

The standardized partial regression coefficient (beta) of FIM-M score and age was -0.561 and 0.027 , respectively.

DISCUSSION

The addition of ME to our ordinary stroke rehabilitation program resulted in increases in the CSA of the thigh both on the affected and unaffected sides, as indicated by differences in CSA between the two patient groups. The degree of increase was most prominent in patients with low ADL scores at the beginning of ME administration. Side effects of ME administration were not serious and resolved after discontinuation of administration. Subjects' age, start of the experimental period from stroke onset, initial FIM-M score, and initial CSA did not differ significantly between the ME and CT groups. Therefore, the results of the comparison between the two groups would not be influenced by factors other than whether ME was administered.

We must describe the system of stroke rehabilitation in Japan, which may differ from that in other countries primarily because of the national insurance system. For example, stroke patients stay longer in an acute care hospital in Japan than in the United States. In 2009, the average number of days from stroke onset to transfer to a kaifukuki rehabilitation ward and the average length of stay in such a ward were 31 and 72 days, respectively, according to information from a questionnaire that was completed by approximately 60% of the kaifukuki rehabilitation wards in Japan.¹⁴ The

TABLE 2 Significant differences between various time points

	2 wks	4 wks	6 wks
Metenolone enanthate group, affected side			
0 wks	^a	^a	^a
2 wks		^a	^a
4 wks			^a
Metenolone enanthate group, unaffected side			
0 wks	^a	^a	^a
2 wks		^a	^a
4 wks			^a
Control group, affected side			
0 wks	NS	NS	NS
2 wks		NS	NS
4 wks			NS
Control group, unaffected side			
0 wks	^a	^a	^a
2 wks		^a	NS
4 wks			NS
Metenolone enanthate Group vs. Control Group, Affected Side and Unaffected Side			
0 wks	2 wks	4 wks	6 wks
NS	NS	NS	^a

NS, not significant.

^a $P < 0.05$.

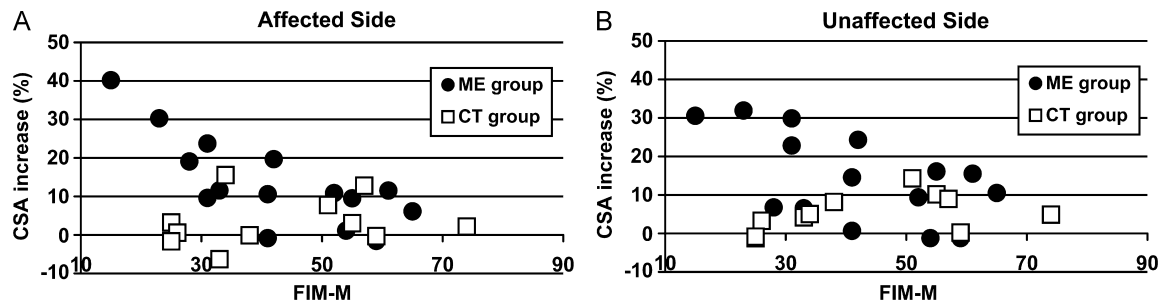


FIGURE 2 A–B, Scattergram of the motor subscore of the Functional Independence Measure (FIM-M) and the increase in the cross-sectional area (CSA) of the thigh muscle in the metenolone enanthate (ME) group and the control (CT) group. Patients with a low initial FIM-M score tended to have a high CSA increase in both sides.

medical insurance system in Japan allows stays of this length. In an acute care hospital, the dose of rehabilitation is often limited to within an hour, and the content tends to focus on range of motion or seated exercises. Therefore, some patients have developed disuse atrophy before they come to a kaifukuki rehabilitation ward.

The use of AS in the field of rehabilitation is rather rare. Bakhsi et al.¹⁵ administered testosterone enanthate to 15 patients who were admitted to a geriatric evaluation and management unit for rehabilitation and found a significant increase in the FIM score and grip power. Kawahira⁸ reported the favorable effect of ME in 25 stroke patients. Improvement in muscle strength was found to be significantly greater after muscle-strengthening exercises combined with AS administration than after muscle-strengthening exercises alone.^{7,8,16} Because vigorous strengthening exercise induces androgen binding sites,¹⁷ a combination of AS and strengthening exercise would seem to be a reasonable treatment strategy. However, patients with disuse atrophy often cannot tolerate vigorous exercise. In this study, ME was administered under conditions of routine rehabilitation without the addition of vigorous exercise. Our favorable results, which were achieved without vigorous exercise, indicate that ME administration could be useful in fragile patients with severe disuse syndrome.

In this study, the patients with lower ADL scores had a greater improvement in CSA than did those with higher ADL scores. This may be inconsistent with the fact that when muscle-strengthening exercise is combined with ME administration, the effect of ME would be enhanced. Therefore, we must consider the situation of the study patients before admission to our hospital. Typically, rehabilitation is insufficient for many stroke patients in acute care hospitals. Patients having a low ADL score on admission to our hospital tend to be very inactive in

the acute care hospital because of their inability to move by themselves and thus easily become deconditioned and even malnourished. The lower the ADL score is, the more severe the degree of deconditioning or malnutrition. This is supported by the results of a study by Kondo and Ota¹⁸ that measured the CSA of leg muscles in 20 stroke patients and found that patients who needed assistance with walking had a smaller CSA than did those who could walk without help. Therefore, our results showing the greatest improvement in those with low initial ADL scores might be explained by the fact that there would be more room for improvement in those patients with a greater degree of disuse atrophy.

Age is a consideration in our study, as stroke patients tend to be aged. The effect of AS on the elderly has been studied.⁴ Urban et al.¹⁹ showed that the strength of the hamstrings and quadriceps of healthy people with an average age of 67 yrs increased with testosterone administration. However, Synder et al.⁵ found no difference in knee extension and flexion strength with the use of a testosterone patch compared with use of a placebo patch in healthy elderly persons older than 65 yrs. The difference in these findings indicates that the effect of AS or testosterone in the healthy elderly is still controversial. Our regression analysis showed a high beta value for FIM-M and a low beta value for age, which indicates that the contribution of age to the CSA change was small. However, there was little variation in age among our subjects.

In general, there is evidence that the effect of AS may be limited in women. The mechanism of this sex difference is explained by the lower antiproteolytic effect of insulin, fewer circulating androgens, and more androgen receptors after AS administration in women than in men.¹⁷ That clinical reports concerning sex differences in AS administration are rare is partially because studies of AS

have been mainly concerned with men. Although we did not aim to eliminate women as study subjects, only three women were included in the ME group. Further studies with a larger number of female patients will be needed to clarify the issue of AS administration in women.

This study has some limitations. The number of subjects was small, and subjects were not blinded as to whether they received ME. However, assignment was random, and subjects had agreed to participate whether they would be placed in the ME or the CT group. Therefore, we did not think it was necessary to use a placebo injection in the CT group. However, in interpreting the superiority of results in the ME group, it should be considered possible that knowledge of receiving ME administration would lead subjects in the ME group to augment daily activity and result in a favorable outcome.

CONCLUSION

Fifteen stroke patients in our kaifukuki rehabilitation ward were administered ME once a week for 6 wks, whereas the 11 patients in the CT group received no such administration. The increase in the CSA of the thigh muscles was larger in the ME group than in the CT group, indicating that ME administration may be an effective treatment in stroke rehabilitation, especially in those with lower ADL scores, as the effect of AS was, to some extent, inversely dependent on the degree of activity in our study patients.

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